

WEEK 7 TASK – WIRESHARK & NETWORK ANALYSIS REPORT

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QUESTION 1: ANALYSE PACKET1.PCAP AND FIND THE FLAG

Analysis Method

The packet capture file packet1.pcap was analysed using Wireshark. During the analysis, ICMP traffic was inspected to identify any unusual packet activity. A suspicious ICMP packet containing encoded data was discovered within the payload section. Further investigation was conducted to determine whether the payload contained hidden information.

Step 1: Identify the Suspicious ICMP Packet

The packet capture file was opened using Wireshark. To locate the suspicious packet, the display filter frame.number == 37 was applied. The selected packet was identified as an ICMP Echo Reply packet containing an unusual payload.

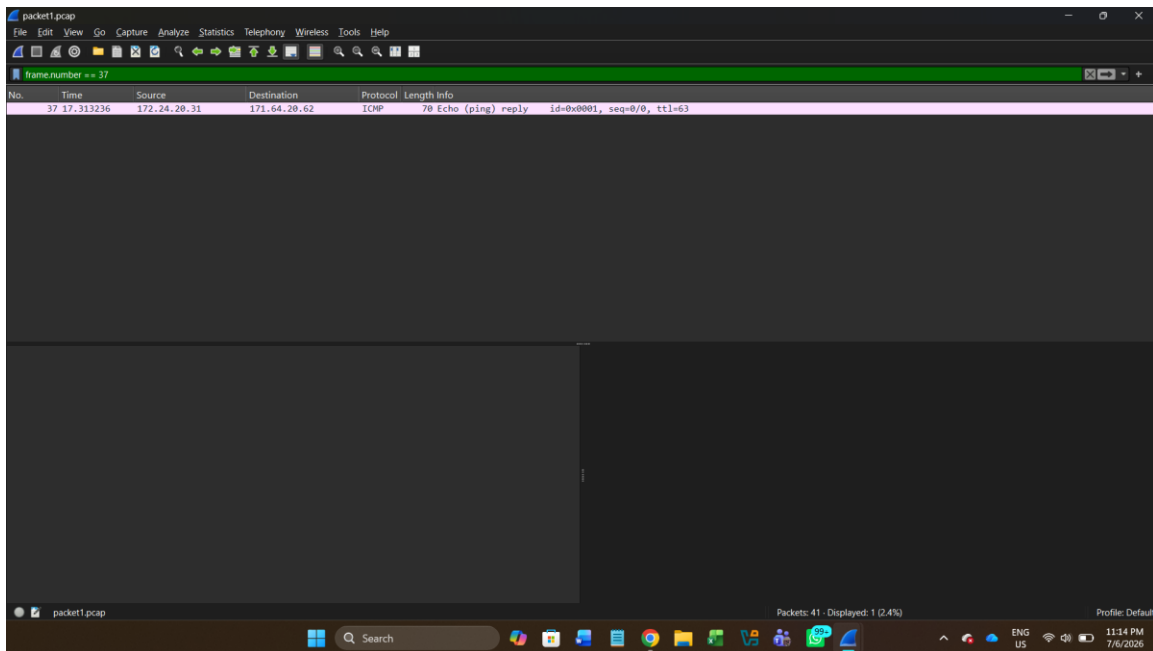


Figure 1: Suspicious ICMP packet containing encoded data in the payload section.

Step 2: Extract and Decode the Payload

The encoded payload was extracted and analysed further. The following encoded string was identified:

U1VDVEYyMDIze2FpX2lzX2Nvb2x9

The format suggested Base64 encoding. CyberChef was used to decode the data.

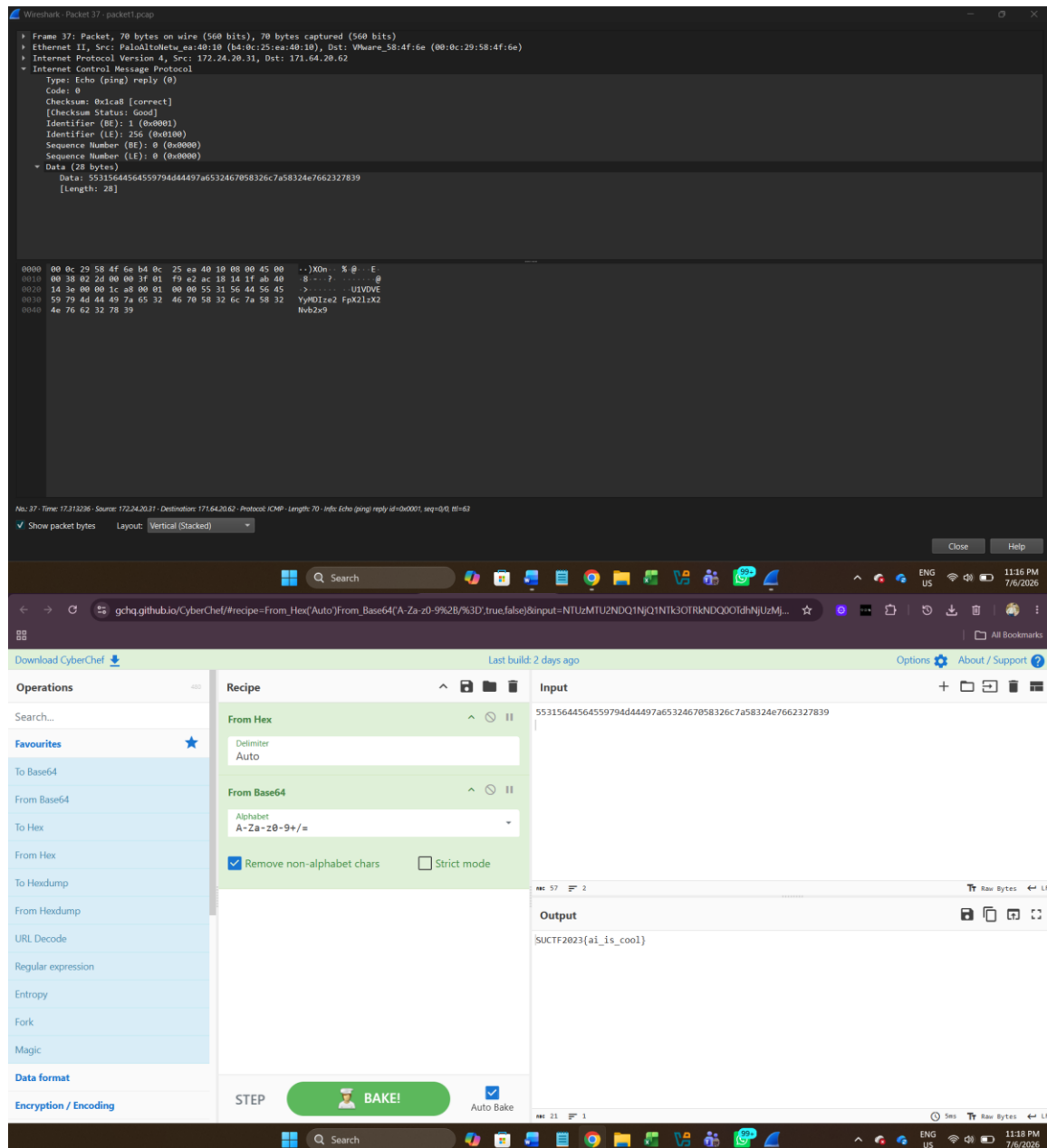


Figure 2: Decoding the extracted payload using CyberChef.

Result

FLAG 1: SUCTF2023{ai_is_cool}

Conclusion

This analysis demonstrated how hidden information can be embedded within network traffic. By examining ICMP packets and decoding the extracted payload, the hidden flag was successfully identified.

QUESTION 2: ANALYSE PACKET2.PCAP AND FIND THE FLAG

Analysis Method

The packet capture file packet2.pcap was analysed using Wireshark. During the investigation, FTP communication was discovered between the client and server. Further analysis revealed that a file transfer had taken place and that the transferred file contained hidden information.

Step 1: Analyse FTP Traffic

The FTP session revealed that an anonymous user had successfully logged into the FTP server and downloaded a file named global_thermonuclear_war.gamerules.txt.

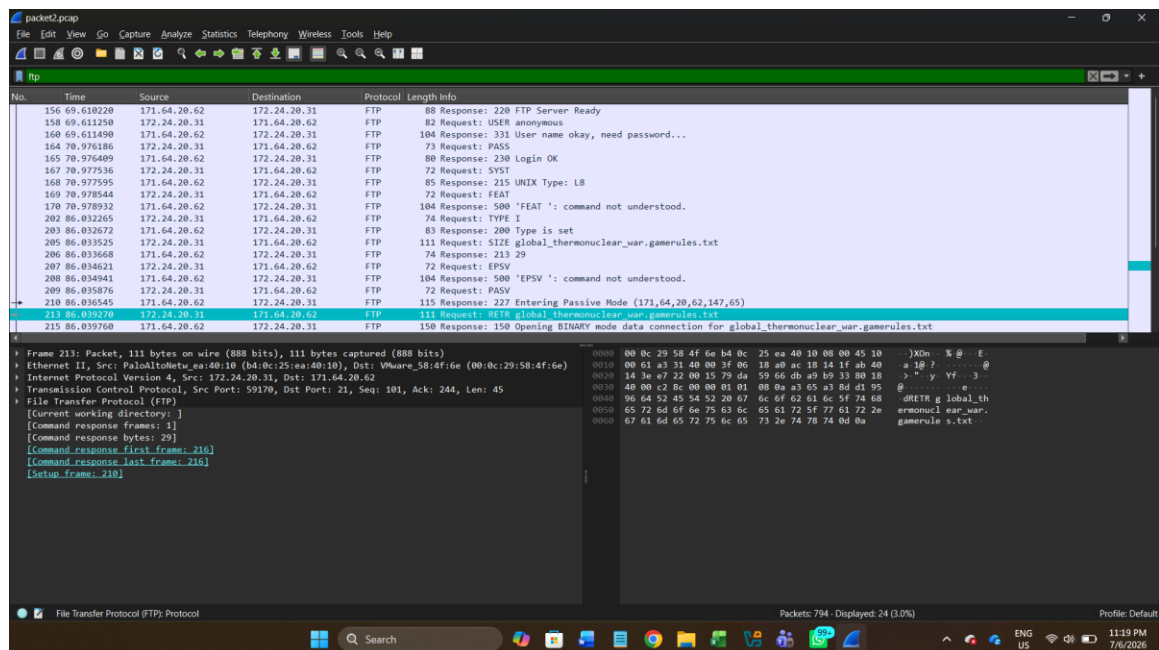


Figure 3: FTP communication showing anonymous login and file download activity.

Step 2: Analyse FTP-DATA Traffic

The FTP-DATA packet associated with the file transfer was located and examined.

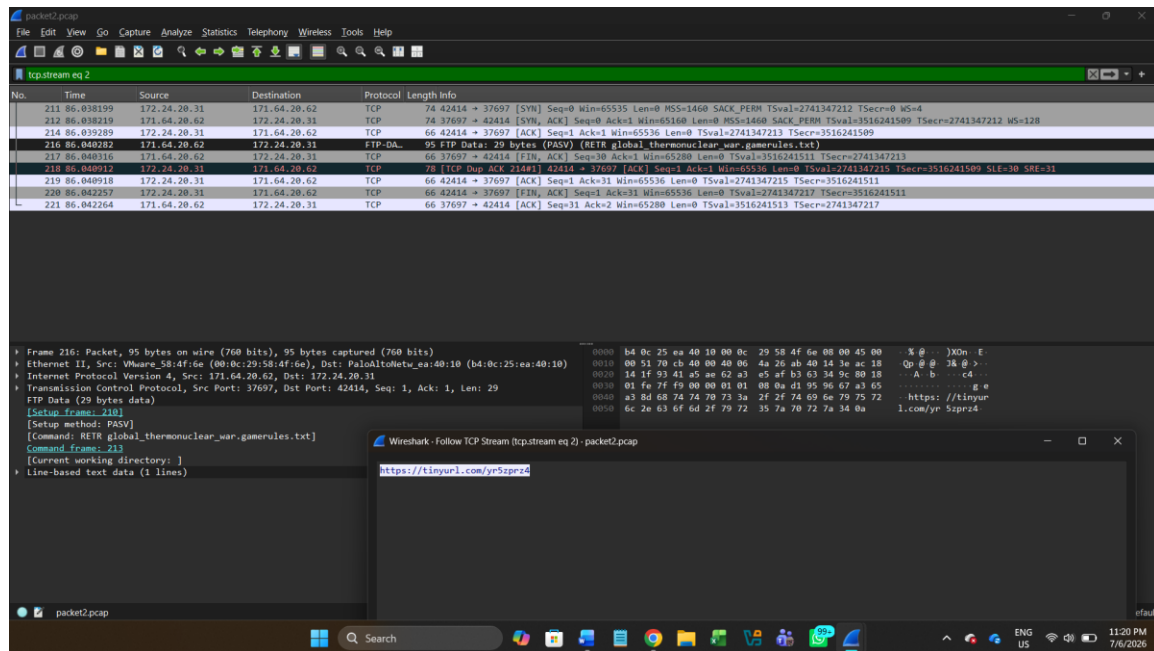


Figure 4: FTP-DATA packet containing file transfer content.

Step 3: Follow TCP Stream

The TCP stream revealed the following URL:

<https://tinyurl.com/yr5zprz4>

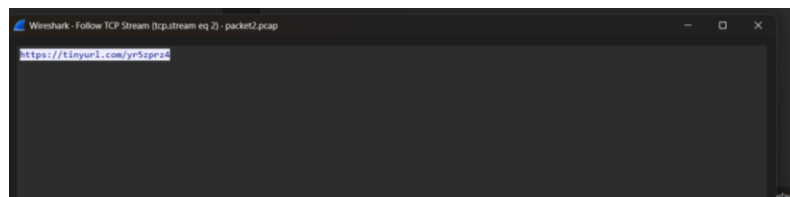


Figure 5: TCP Stream revealing a hidden TinyURL link.

Step 4: Analyse the External Content

The URL was opened in a web browser and redirected to a Google document containing encoded symbols.

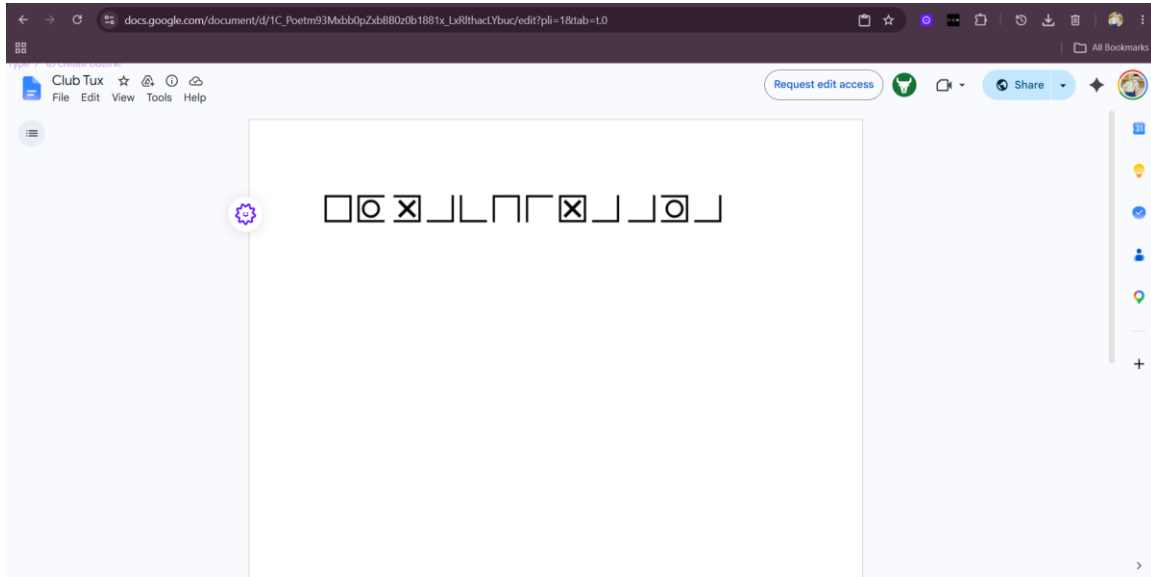


Figure 6: Google document containing encoded symbols.

Result

FLAG 2: SUCTF2023{EX MACHINA AVA}

Conclusion

By analysing FTP traffic, extracting transferred data, and following the TCP stream, hidden information was successfully recovered.

REFERENCES

1. Wireshark Foundation – <https://www.wireshark.org>
2. CyberChef – <https://gchq.github.io/CyberChef>
3. TinyURL – <https://tinyurl.com/yr5zprz4>
4. Google Docs – Online document platform used during the investigation.